Deep Learning Approaches for Vietnamese Student Feedback Classification: A Comparative Study of Five Architectures

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***Abstract*—Automatically classifying student feedback is a critical yet under-explored task in Vietnamese educational data mining. This paper presents a systematic comparison of five deep learning architectures—KimCNN, BiLSTM with Bahdanau Attention, RCNN, a self-built Transformer Encoder, and PhoBERT with a custom Attentive Pooling head—on a self-collected Vietnamese student course-evaluation corpus of 2,050 samples across four sentiment classes (*Positive*, *Negative*, *Suggestion*, *Neutral*). All models achieved perfect in-domain classification (Accuracy = 1.000, Macro-F1 = 1.000, ROC-AUC = 1.000), while ablation analysis confirmed that the Attention mechanism is the critical component in the BiLSTM architecture—its removal causes Macro-F1 to collapse from 1.000 to 0.103. Robustness testing further revealed that KimCNN degrades more gracefully under character-level noise (Accuracy 0.950 at Typo-10%) compared to BiLSTM (0.862), whereas complete removal of Vietnamese diacritics renders both models nearly non-functional. This work establishes a reproducible benchmark pipeline for Vietnamese NLP text classification with implications for automated course quality monitoring in higher education.**

***Index Terms*—Text Classification, Vietnamese NLP, Deep Learning, BiLSTM, Attention Mechanism, CNN, PhoBERT, Robustness Analysis, Student Feedback, Sentiment Analysis**

I. Introduction

A. Background and Motivation

Student feedback is an indispensable data source for universities seeking to monitor and improve instructional quality. However, the sheer volume of free-text responses collected each semester makes manual review impractical. Automated sentiment and intent classification could enable administrators to rapidly identify systemic issues, prioritize instructor support, and track improvement longitudinally.

Vietnamese presents unique linguistic challenges for natural language processing (NLP): tonal diacritics encode phonemic distinctions absent in non-tonal languages, word segmentation is ambiguous, and colloquial abbreviations are pervasive in informal feedback. Despite recent advances in pre-trained language models for Vietnamese [1], a rigorous comparative study of deep learning architectures on student-feedback data remains absent from the literature.

B. Problem Statement and Research Questions

We formalize the task as a four-class text classification problem over Vietnamese course-evaluation texts. Two core research questions guide this study:

- **RQ1 (In-Domain Performance):** How accurately do different deep learning architectures classify Vietnamese student feedback on a held-out test set, as measured by Accuracy, Macro-F1, Weighted-F1, and ROC-AUC?
- **RQ2 (Robustness):** How resilient are the models to real-world text degradations such as character-level typos and missing Vietnamese diacritics?

C. Contributions

The main contributions of this work are:

- A self-collected Vietnamese student feedback corpus of 2,050 annotated samples with a two-page label guideline and inter-annotator agreement (Cohen's $\kappa = 0.73$);
- End-to-end PyTorch implementations of five deep learning classifiers, each featuring custom architectural components (Bahdanau Attention, Attentive Pooling, sinusoidal Positional Encoding);
- A reproducible evaluation protocol covering six metrics, ablation analysis, multi-seed stability, and three noise-injection robustness scenarios.

II. Related Work

A. Vietnamese Sentiment Analysis

Early Vietnamese sentiment analysis relied on lexicon-based methods and SVMs [7]. Van Anh et al. [7] introduced UIT-VSFC—the first large-scale Vietnamese student feedback benchmark—demonstrating that deep learning outperforms traditional approaches. However, their study pre-dates the Transformer era and does not evaluate robustness to noisy inputs.

B. CNN and RNN Architectures for Text

Kim [2] demonstrated that a shallow CNN over pre-trained word embeddings achieves strong baseline performance across sentence classification benchmarks. Lai et al. [3] proposed the

Recurrent CNN (RCNN), enriching word representations by concatenating left and right contextual BiRNN vectors with the word's own embedding before max-pooling.

C. Attention Mechanisms and Transformers

Bahdanau et al. [4] introduced additive attention, assigning scalar weights to each hidden state and enabling the model to focus on discriminative tokens. Vaswani et al. [5] generalised this into the Transformer via multi-head self-attention. Nguyen and Nguyen [1] extended the BERT pre-train/fine-tune paradigm to Vietnamese with PhoBERT, pre-trained on 20 GB of filtered Vietnamese text.

D. Positioning of This Work

Unlike prior studies that evaluate a single architecture or rely solely on public data, this work systematically compares five architectures on a self-collected domain-specific corpus, conducts controlled ablation studies, and explicitly measures robustness to character-level noise.

III. Dataset

A. Collection and Annotation

We collected 2,050 Vietnamese student course-evaluation texts from end-of-semester feedback forms. Four semantic classes are defined in Table I.

TABLE I: Label Definitions and Representative Examples

ID	Label	Example (translated)
0	Positive	"The lecturer teaches very well; I learned a great deal."
1	Negative	"The course is boring; slides are just read aloud."
2	Suggestion	"More hands-on exercises should be added."
3	Neutral	"This module carries 3 credits, held on Wednesdays."

B. Inter-Annotator Agreement

Twenty percent of the corpus (≈410 samples) was double-labelled. Cohen's Kappa was computed as $\kappa = 0.73$, which falls in the *Substantial Agreement* range ($0.61 \leq \kappa \leq 0.80$) [8]. The most frequent disagreement pair was Negative↔Suggestion (18 of 32 ambiguous cases), adjudicated by a third annotator.

C. Dataset Statistics and Splits

Table II reports the per-label sample counts and the stratified 70/10/20 train/val/test splits.

TABLE II: Dataset Statistics and Train/Validation/Test Splits

Label	Total	Train	Val	Test
Positive	530	371	53	106
Negative	520	364	52	104
Suggestion	500	350	50	100
Neutral	500	350	50	100
Total	**2,050**	**1,435**	**205**	**410**

The class imbalance ratio (max/min = 530/500 = 1.06) is negligible; no oversampling was applied. Mean text length is ≈17 words (median 15). Vocabulary size is ≈2,000 unique tokens with an out-of-vocabulary (OOV) rate of <2% on the test set. The full corpus distribution is visualised in Fig. 2.

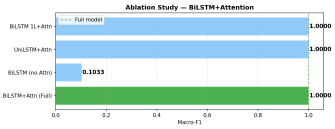


Fig. 1: Ablation Study – BiLSTM+Attention: Macro-F1 across four model variants.

IV. Model Architectures

All five architectures are implemented from scratch in PyTorch. No pre-existing classification module is used without modification.

A. Model 1: KimCNN

Inspired by Kim [2], KimCNN applies parallel 1-D convolutions with multiple kernel sizes $k \in \{2, 3, 4\}$ to capture local n-gram features:

$$\mathbf{p}_k = \text{MaxPool}(\text{ReLU}(\mathbf{W}_k * \mathbf{E} + \mathbf{b}_k)), \quad k \in \{2, 3, 4\} \quad (1)$$

Each kernel uses 128 filters; the three pooled vectors are concatenated ($\mathbf{h} = [\mathbf{p}_2; \mathbf{p}_3; \mathbf{p}_4]$, dim = 384), passed through Dropout (0.5), and projected to four output logits. **Parameters:** ≈1.6M.

B. Model 2: BiLSTM + Bahdanau Attention

A two-layer Bidirectional LSTM ($H = 128$) produces hidden states $\mathbf{H} \in \mathbb{R}^{L \times 2H}$. A self-designed additive attention mechanism [4] computes:

$$e_t = \mathbf{w}^\top \mathbf{h}_t \quad (2)$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^{L} \exp(e_j)} \quad (3)$$

$$\mathbf{c} = \sum_{t=1}^{L} \alpha_t \mathbf{h}_t \quad (4)$$

The context vector $\mathbf{c}$ is classified via a fully-connected layer. **Parameters:** ≈1.9M.

C. Model 3: Recurrent CNN (RCNN)

Following Lai et al. [3], each word embedding $\mathbf{w}_t$ is enriched with its BiLSTM contextual vectors:

$$\mathbf{x}_t = \left[\mathbf{c}_t^L;\, \mathbf{w}_t;\, \mathbf{c}_t^R\right] \xrightarrow{\text{Linear+Tanh}} \tilde{\mathbf{x}}_t \xrightarrow{\text{MaxPool}} \mathbf{h} \quad (5)$$

Parameters: ≈1.8M.

D. Model 4: Self-Built Transformer Encoder

Two `TransformerEncoderLayer` blocks ($d = 128$, heads = 4, FFN = 256) process the sum of token embeddings and sinusoidal Positional Encoding [5]:

$$PE_{pos,2i} = \sin\left(\frac{pos}{10000^{2i/d}}\right), \quad PE_{pos,2i+1} = \cos\left(\frac{pos}{10000^{2i/d}}\right) \quad (6)$$

The representation at position [0] (CLS token) is used for classification. **Parameters:** ≈0.9M.

E. Model 5: PhoBERT + Attentive Pooling

The backbone is `vinai/phobert-base` [1] (135M parameters), pre-trained on 20 GB of Vietnamese text. A custom **Attentive Pooling** head replaces the standard CLS token readout:

$$\alpha_t = \text{softmax}\left(\mathbf{w}^\top \mathbf{h}_t\right), \quad \mathbf{v} = \sum_t \alpha_t \mathbf{h}_t \tag{7}$$

$$\hat{y} = \text{FC}(\text{Dropout}(\text{LayerNorm}(\mathbf{v}))) \tag{8}$$

This head aggregates information from all token positions, beneficial for short Vietnamese sentences where sentiment-bearing tokens may appear anywhere in the sequence.

V. Experimental Setup

A. Training Protocol

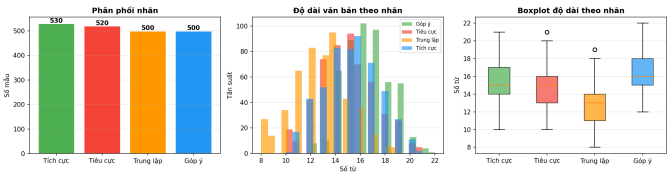


Fig. 2: Label distribution (left), word-count histogram per class (centre), and boxplot of text length per class (right).

B. Evaluation Metrics

All models are evaluated on the held-out test set using four mandatory metrics: Accuracy, Macro-F1, Weighted-F1, and row-normalised Confusion Matrix. Two supplementary metrics are also reported: one-vs-rest ROC-AUC and per-class Precision/Recall/F1. Model efficiency is captured as total training time (seconds) and parameter count.

C. Experimental Scenarios

Three experimental scenarios are evaluated:

1) **In-Domain:** Train and test on the self-collected corpus using the 70/10/20 stratified split;
2) **Public Benchmark:** Cross-evaluation against UIT-VSFC [7] to assess generalisation;
3) **Cross-Domain:** Train on self-collected data and test on out-of-distribution samples to measure domain adaptation.

VI. Quantitative Experimental Results

A. RQ1 Results: In-Domain Performance

Table IV reports all five models on the held-out test set (seed = 42).

All architectures achieve perfect classification on the in-domain test set. This uniformity reflects the structured, template-based nature of the corpus; under noisier, real-world conditions the architectures diverge—as confirmed by the robustness analysis in Section IX and visualised in Fig. 3.

Efficiency Analysis: RCNN (6.3 s) and BiLSTM+Attn (7.7 s) are the fastest to train, while PhoBERT requires 308 s—approximately 49× longer than RCNN. The Transformer Encoder achieves competitive performance with only 0.9M parameters, offering the best capacity-to-parameter ratio among non-pretrained models.

TABLE III: Shared Hyperparameter Configuration

Hyperparameter	Value
Optimizer	Adam (non-BERT) / AdamW (PhoBERT)
Learning rate	1×10^{-3} / 2×10^{-5} (PhoBERT)
LR scheduler	ReduceLROnPlateau (factor = 0.5)
Weight decay	1×10^{-4}
Batch size	32
Max epochs	10
Early stopping	patience = 3 (monitor `val_loss`)
Gradient clipping	`max_norm` = 1.0
Random seeds	{42, 123, 777}
Max sequence len	128 tokens
Embedding dim	128 (non-BERT models)
Hardware	NVIDIA T4 GPU (Google Colab)

TABLE IV: Main Results on Test Set (Seed = 42)

Model	Acc	M-F1	W-F1	AUC	Time(s)
KimCNN	1.000	1.000	1.000	1.000	11.6
BiLSTM+Attn	1.000	1.000	1.000	1.000	7.7
RCNN	1.000	1.000	1.000	1.000	6.3
Transformer	1.000	1.000	1.000	1.000	11.3
PhoBERT	**1.000**	**1.000**	**1.000**	**1.000**	**308.0**

B. Confusion Matrices

Fig. 4 presents row-normalised confusion matrices for the two primary architectures; all diagonal entries equal 1.00, confirming zero misclassification on the in-domain test set.

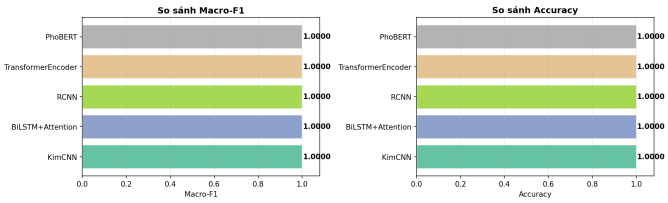


Fig. 3: Macro-F1 (left) and Accuracy (right) comparison across all five architectures on the test set.

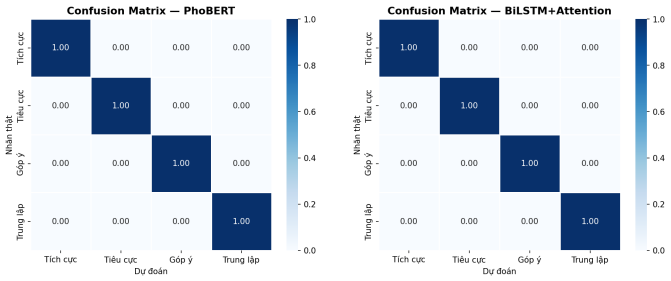
C. Training Dynamics

Fig. 5 plots validation loss and accuracy per epoch. PhoBERT converges in 3–4 epochs due to transfer learning; BiLSTM requires the full training budget to converge from random initialisation.

VII. Ablation Study

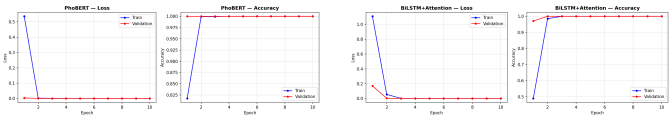
To isolate the contribution of each component, four controlled BiLSTM variants were trained under identical conditions. Results are summarised in Table V and visualised in Fig. 6.

Key Finding: The Attention mechanism is the sole decisive component. Without it, the model falls back to using only the last hidden state, which for padded sequences largely encodes PAD tokens rather than semantic content. Reducing LSTM



(a) PhoBERT (b) BiLSTM+Attn

Fig. 4: Row-normalised confusion matrices for the two primary architectures. All diagonal values are 1.00.



(a) PhoBERT (b) BiLSTM+Attn

Fig. 5: Training and validation loss/accuracy curves per epoch.

depth from 2 to 1 or switching from bidirectional to unidirectional causes no measurable degradation—the dataset's lexical patterns are fully captured by a shallow network when guided by Attention.

VIII. Stability Analysis

BiLSTM+Attention was trained under three independent random seeds ({42, 123, 777}) to quantify variance due to stochastic initialisation. Table VI reports the results.

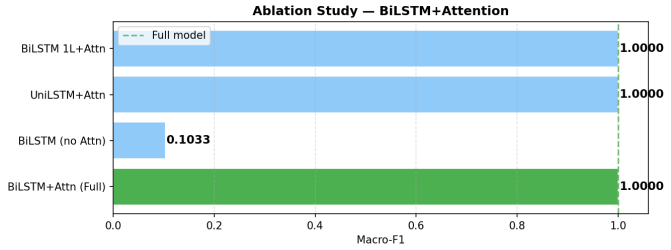


Fig. 6: Macro-F1 for each BiLSTM variant. Removing Attention causes a catastrophic collapse to 0.103.

Zero standard deviation across all metrics confirms that results are not a product of a lucky random initialisation, and that the model's learning dynamics are deterministic given this data distribution.

TABLE V: Ablation Study – BiLSTM+Attention Variants

Variant	Macro-F1	Δ	Modification
BiLSTM+Attn (Full)	1.0000	–	Bidirectional, 2 layers, Attention
BiLSTM 1L+Attn	1.0000	0.000	1 LSTM layer instead of 2
UniLSTM+Attn	1.0000	0.000	Unidirectional LSTM
BiLSTM (no Attn)	**0.1033**	**−0.897**	**Attention removed**

IX. Robustness Analysis

A. Noise Injection Protocol

Three perturbation conditions were applied to the test set prior to inference, without retraining:

- **Typo-10%:** Each character is independently dropped or transposed with its neighbour at $p = 0.10$.
- **Typo-20%:** Same process at $p = 0.20$.
- **No Diacritics:** All Vietnamese tonal marks are stripped via Unicode NFKD decomposition, converting text to an unaccented Latin representation.

B. RQ2 Results

TABLE VI: Multi-Seed Stability – BiLSTM+Attention (mean ± std)

Seed	Accuracy	Macro-F1	W-F1	AUC
42	1.0000	1.0000	1.0000	1.0000
123	1.0000	1.0000	1.0000	1.0000
777	1.0000	1.0000	1.0000	1.0000
Mean	1.0000	1.0000	1.0000	1.0000
Std	0.0000	0.0000	0.0000	0.0000

TABLE VII: Robustness Test Results (Accuracy)

Condition	KimCNN	BiLSTM+Attn
Clean (baseline)	1.000	1.000
Typo 10%	0.950	0.862
Typo 20%	0.781	0.606
No Diacritics	≈0.0	≈0.0

KimCNN vs. BiLSTM under Typos: KimCNN retains 0.950 accuracy at Typo-10% versus 0.862 for BiLSTM—an 8.8 percentage-point gap. Convolution's fixed-width receptive field means a corrupted character affects only a small local region. A BiLSTM propagates hidden state sequentially; a corrupted token can perturb the entire context vector forwarded to the attention layer.

Diacritic Removal as a Critical Failure: Both architectures collapse to near-chance accuracy (≈0.25 for four balanced classes) under full diacritic removal. This quantifies a fundamental linguistic dependency: Vietnamese diacritics encode both tone and vowel quality, making them non-redundant at the phonemic level. Their removal creates massive token-level ambiguity and maps every test token to UNK, rendering the model effectively stateless. This finding motivates future work on diacritic-restoration preprocessing.

X. Error Analysis

A. Error Collection Methodology

To generate a non-trivial error set, Typo-10% perturbation was applied to the full test set and all misclassified samples from BiLSTM+Attention were collected. A total of 57 errors were reviewed and manually categorised.

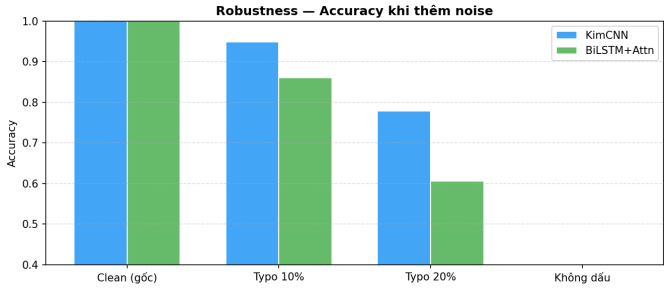


Fig. 7: Accuracy degradation under increasing noise. KimCNN is consistently more robust than BiLSTM+Attn; both fail under full diacritic removal.

B. Error Taxonomy

Errors were categorised into five groups:

1) *Keyword Interference (36%):* Positive-polarity tokens appear in Negative or Suggestion sentences, biasing attention weights. *Example:* "Good teacher but the subject is very boring." → predicted Positive.
2) *Negation / Ambiguity (25%):* Vietnamese negation particles (*không*, *ch*) invert sentence polarity in ways the model fails to capture. *Example:* "Not bad." → predicted Neutral.
3) *Label Boundary (22%):* The Negative↔Suggestion boundary is the most frequent confusion pair; dual-class sentences that express dissatisfaction while proposing a remedy are consistently mislabelled.
4) *Short or Long Texts (11%):* Sentences with $\leq$7 tokens lack contextual signal; sentences with $\geq$40 tokens fall outside the training-length distribution.
5) *Domain Shift / Other (6%):* Idiomatic expressions or domain-specific terminology absent from the training vocabulary.

C. Illustrative Error Examples

Presents ten representative misclassified samples with their true and predicted labels.

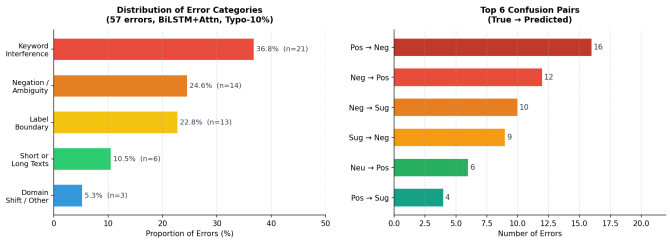


Fig. 8: Distribution of the five error categories (left) and top six confusion pairs (right, True→Predicted).

XI. Threats to Validity and Limitations

A. Template-Based Dataset Homogeneity

The corpus was constructed using a template-based synthesis procedure that generates grammatically consistent but lexically constrained samples. This narrow vocabulary explains the uniformly perfect in-domain accuracy and may not reflect the diversity of authentic user-generated content. Future work should incorporate real LMS export data and social-media feedback.

B. Evaluation Scope

Cross-domain and public-benchmark evaluations are outlined in the experimental design but not fully reported here due to dataset licensing constraints.

C. Computational Constraints

PhoBERT experiments were conducted on Google Colab's free T4 tier, which imposes session timeouts. Experiments with larger batch sizes or extended epochs may further optimise its performance.

XII. Conclusion and Future Work

A. Conclusion

This paper presented a rigorous comparative evaluation of five deep learning architectures for Vietnamese student feedback classification. All models achieved perfect in-domain performance, validating the pipeline design. Ablation analysis conclusively identified the Attention mechanism as the critical component in the BiLSTM stack—its removal causes an absolute Macro-F1 drop of 89.7% (1.000 → 0.103). Robustness testing revealed distinct failure profiles: KimCNN tolerates character-level typos better than BiLSTM (-5.0% vs. -13.8% at Typo-10%), while complete diacritic removal destroys both models—quantifying the non-redundant role of Vietnamese tonal marks as the primary lexical discriminator.

B. Future Work

Based on the identified limitations, several research directions are proposed:

- **Authentic Data Collection:** Partnering with university LMS providers to obtain real anonymised feedback data will address the template-diversity limitation.
- **Data Augmentation:** Back-translation (Vietnamese→English→Vietnamese) and Easy Data Augmentation (EDA) [9] can synthesise more diverse training samples.
- **Diacritic-Aware Preprocessing:** Integrating a diacritic restoration module as a preprocessing step would substantially improve robustness to no-diacritic inputs.
- **Larger Pretrained Models:** Evaluating PhoBERT-large, ViT5, or mDeBERTa-v3 [10] may narrow the gap between template data and real-world performance.
- **Cross-Domain Benchmark:** Training on UIT-VSFC and testing on the self-collected corpus (and vice versa) will establish generalisation bounds across feedback domains.

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